SIEMENS Healthineers

MAGNETOM

Troubleshooting Guide System

Patient Handling

Harmony syngo MR
Symphony syngo MR
Sonata syngo MR
Trio
Upgrades to syngo MR
from Impact/Expert
from Vision

Document Version

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1.1 General

The main tool for patient handling troubleshooting are:

- PT communication and status test is available in the service software platform "Test Tools" and the Service Software for Patient Table (see PT Procedure).
- The error messages shown in the Event Log. PMU communication and status test, available in the service software platform "Test Tools".

The (→ FRU Overview / MR-000.841.15) lists the necessary adjustments and final checks when replacing a FRU.

For specific error messages, check the error message in System/Report/Detail for the corrective action.

A detailed describtion about the error is part of the error messages; all message ID's are added in a table in this TSG.

> If a fatal error occurs, the funnel symbol on the magnet display will begin flashing.

(→ Description of LED's / Page 36) and (→ Description of fuses / Page 38) are located in the (→ Procedure / Page 26)

1.2 General troubleshooting hints

Feed back from the field and the manufacturer of the patient table has shown a high spare part consumption without solving the problem, e.g. 8**0% of the returned controllers A5140/A514S are not defective**.

The following hints should facilitate troubleshooting and prevent from unnecessary part replacement.

> Before exchanging any electronic components of the P-Tab check the proper vertical/ horizontal drive conditions.

1.2.1 Vertical movement problems

Essential for the correct functioning of the vertical drive are:

- Proper adjustment of the vertical switches (SZ10, SY0...SY5)
- Regular maintenance of the spindle grease cartridge, refer to SI 17/04 and maintenance instruction. If the grease cartridge is not replaced once per year it may happen that the vertical drive switches-off with over current sporadically. The current value can be monitored by looking at the digital display of the converter A511 (Micro master). The maximum current (200Kg load) should not exceed 6A.
- Proper spindle alignment. If the spindle is not properly aligned, e.g. after replacement, it may lead to high friction and sporadic switch-off of the vertical drive (due to over current, see above). For spindle adjustment please refer to (→ Adjustment / MR-000.841.15).

1.2.2 Horizontal movement problems

Essential for the correct functioning of the horizontal drive are:

- Proper adjustment of switches SY0, SZ10 and the patient table alignment.

Before replacing the horizontal power stage A512 or the main controller A5140/A514S check the condition of the Control panels (switches and joysticks are exposed to high mechanical stress during daily operation). For control panel checks please refer to this TSG.

1.2.3 General Problems

- Defective patient table electronic components due to insufficient protection against condensation water; see also SI 05/03·
- Sporadic/permanent CAN bus problems; mechanical damage of FOC's due to wrong handling of the side covers, or missing metal protection plate (refer to installation instruction).
- When you suspect a problems with the PMU or the Gradient temperature control ("D9") do not exchange the controller. Although they are in the same rack and on the same transformer, but they control different circuits.

- When you suspect image problems regarding to RF-noise do not exchange the power supply A5131 or A5132. It will not solve the problem.

1.2.4 Important for spare parts handling·

- If a defective patient table electronic component is sent back for repair, the manufacturer needs a detailed error description, preferably the event-log message with corresponding error ID.

1.3 Patient Table

The patient table test consist of communication/status request only.

1.3.1 Communication

The test provides you with a listing of the present status of the selected component, i.e. all possible error/status conditions, which will be recognized and stored by the local application software of the component CAN-unit.

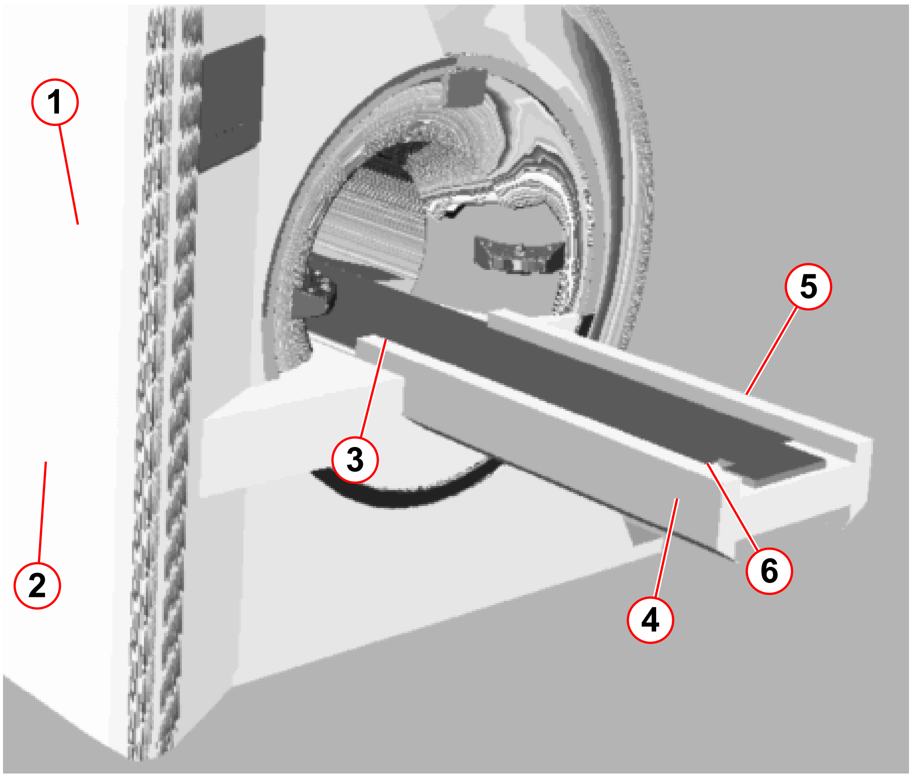
Error conditions are marked with "!!!".

The table below contains a list of the table status messages, their explanations and trouble shooting hints.

> **i** If no PT status is reported, check the CAN bus connection to the patient table controller

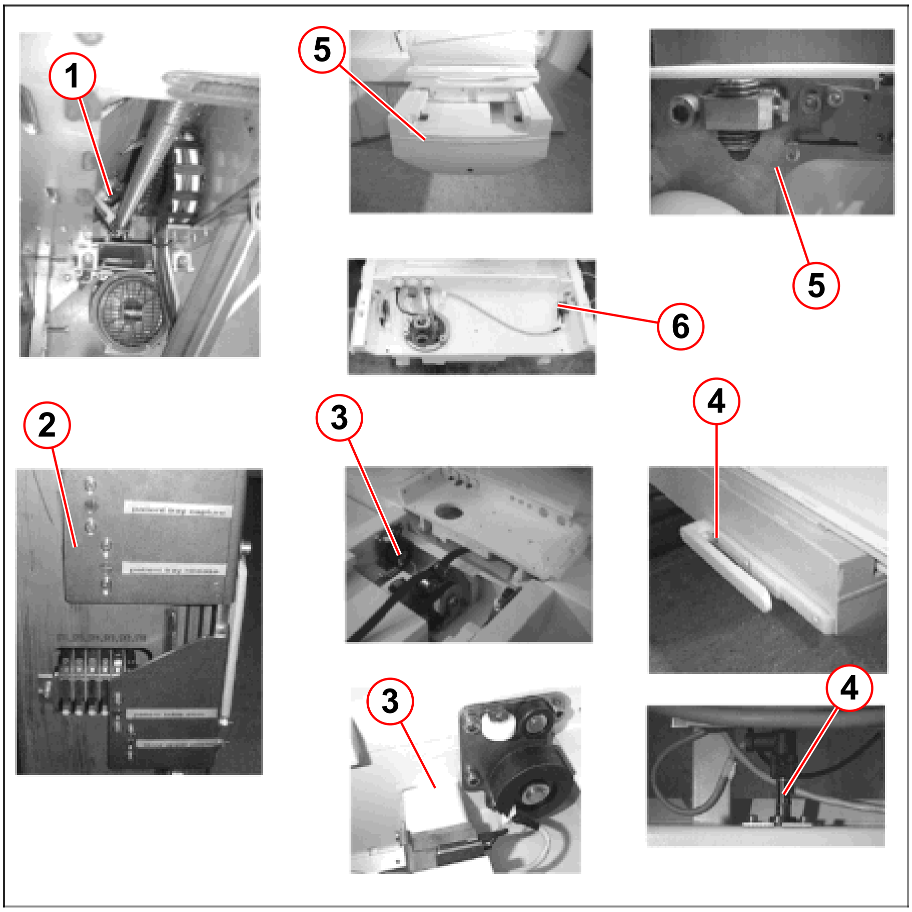
1.3.2 Switch overview (with HaSy magnet covers)

Fig. 1:



(1) SY6
(2) SY1, SY5, SY4, SY3, SY2, SY0
(3) SZ10
(4) SZ12
(5) SZ15a, SZ15b, SZ11
(6) SZ14a, SZ14b, SZ13

Fig. 2: Switch overview



(1) SY6
(2) SY1, SY5, SY4, SY3, SY2, SY0
(3) SZ10
(4) SZ12 (left or right)
(5) SZ15a, SZ15b, SZ11
(6) SZ14a, SZ14b, SZ13 (The switch is a wire bridge)

1.3.3 Switch Status

Status message	Explanation	Status	Hints for troubleshooting
Different sensors	Power supply monitoring for the switches, sensors and table stop circuit	ok	Check fuse F11 on A5132.
+10.5V for cabin UI	Power supply monitor for the control components and display	ok	Check fuses F13 on A5132 and F3 on A5131.
+10.5V for A5181 (PMU)	Power supply monitor for the PMU-converter A5181.	ok	Check fuses F15 on A5132 and F3 on A5131.
+18V for PMU	Power supply monitor for the PMU Front end A5191 and PMU converter A5181.	ok	Check fuses F16 on A5132 and F1 on A5131.
+13.5V for Video	Power supply monitor for the video camera A 5196.	ok	Check fuses F12 on A5132 and F1 on A5131.
-18V for PMU	Power supply monitor for the PMU Front end A5191 and PMU converter A5181.	ok	Check fuses F17 on A5132 and F2 on A5131.
+18V for A5192 (Grad.Tmp.)	Power supply monitor for the gradient temperature supervision on A5192.	ok	Check fuses F18 on A5132 and F1 on A5131.
-18V for A5192 (Grad.Tmp.)	Power supply for the gradient temperature supervision on A5192.	ok	Check fuses F19 on A5132 and F2 on A5131.
For all switch status messages		▪ Check the functionality of the switches using (→ Mode M1, switches and sensors / Page 27). ▪ Check the signal path up to the backplane A5150. ▪ Replace controller A5140/A514S	

Status message	Explanation	Status	Hints for troubleshooting
Switch input Y 0_2	Toggle switch, in case the table is in the vertical mode the horizontal mode is locked.	OFF, when the patient table is in the vertical mode, i.e. the converter is active (the frequency converter display shows 000.0) On, when the patient table is in the horizontal mode	Refer to (→ Switch SY0/ SY2 / Page 31)5
Switch input Y 1	table lower position (without trolley connected)	OFF, when the patient table has reached the lowest position.	
Switch input Y 2	during the table up movement the table speed will be switched to slow speed	OFF, just before the patient table reaches the highest position (height to enter the magnet).	Refer to (→ Switch SY0/ SY2 / Page 31)
Switch input Y 3	If the trolley is connected and the table moves down, the table speed will be switched to slow speed. If the trolley is connected and the table moves up, the table slowly stops	OFF, when the patient table is above the trolley docking height or at the trolley docking height.	Refer to (→ Switch SY3/ SY4 / Page 32) Detailed Adjustment Instructions are located in (→ Adjusting the docking height / M1-010.815.01)
Switch input Y 4	If the trolley is connected and the table moves down, the table slowly stops	OFF, when the patient table is below the trolley docking height or at the trolley docking height.	
Switch input Y 5	Table lower position (with trolley connected)	OFF, when the patient table has reached the lowest position.	
Switch input Y 6_15	Supervision for the spindle nut (switch SY6) Supervision for the carrier panel (switch SZ15a and SZ15b are in series)	not ok, when the spindle nut is out of the guide and/or when the carrier plate is raised up.	Objects may be located underneath the patient table or the carrier plate. Refer to (→ Switch SZ15a/ b / Page 35)
Switch input Y 0_1	Toggle switch, in case the table is in the horizontal mode, the vertical mode is locked.	ON, when the patient table is in the vertical mode, i.e. the converter is active (the frequency converter display shows 000.0). Off, when the patient table is in the horizontal mode.	Refer to (→ Adjusting the docking height / M1-010.815.01)

Status message	Explanation	Status	Hints for troubleshooting
Switch input Y	Checks whether the switch status of switch SYnn is valid.	ok	Refer to (→ Status of switches / Page 18)
Switch input Z	Checks whether the switch status of switch SZnn is valid.	ok	Refer to (→ Status of switches / Page 18)
Switch input Z 10	Locks vertical movement when the table is inside the magnet. Reference position of the Z-axis.	ON, when the table is completely outside the magnet (Home position).	Refer to (→ Switch SZ10 / Page 33)
Switch input Z 11	Sensor; the downward table movement stops, if the trolley is not connected properly or if any obstacle is close to the sensor.	OFF, when an obstacle is located within the range of the sensor. The sensor is not always supplied with voltage. Exception: voltage is always present in service mode.	Refer to (→ Switch SZ11 / Page 34) Check whether obstacles are located underneath the patient table.
Switch input Z 12	Sensor; verifies the trolley connection to the patient table.	ON, when the trolley is docked. The sensor is not always supplied with voltage. **Exception**: voltage is always present in service mode.	Refer to (→ Switch SZ12 / Page 35)
Switch input Z 13	The connection between the carrier panel and the removable tabletop will be verified. The switch is a jumper on connector E543-X2 of the ASV.	OFF, when the carrier plate is moved up.	
Switch input Z 14 a&b	Two switches in parallel; the trolley connection to the carrier panel will be verified	OFF, when the trolley is connected to the carrier panel (i.e. the carrier panel is positioned on the trolley.)	Check that both trolley latches are locked in.
SW cond. 0 to 7	Current status of the software	Not relevant.	

Status message	Explanation	Status	Hints for troubleshooting
Cabin Panel 0	Code in front right operating panel	Present, when the respective operating panel is connected. None, corresponding operating panel is not connected.	Check the connection to the operating panel.
Cabin Panel 1	Code in front left operating panel		
Cabin Panel 2	Code in back right operating panel		
Display 0	Code for front display	Present, when the respective display is connected. None, corresponding display is not connected.	Check the connection to the display.
Display 1	Code for back display		
I2C Power Supply	Reports a malfunction of the I2C bus interface of the A5132 power supply	ok	Power-on-Reset via F15 (PCA)
I2C Bus	Reports a malfunction of the I2C bus system caused by an incorrect function of the operating panel or display.	ok	Ignore the error, if it occurs only once. Replace operating panel A516 or display A517. Replace power supply A5132 if incorrect error messages regarding power failures are displayed. Replace controller A5140/A514S.
Table move	Motorized table movement is enabled.	Not relevant.	
Error output of converter	Internal error of the frequency converter A511.	On	Reload/All. Check whether there are any mechanical problems (table should move smoothly). Overload for table due to non-stop movement or an excessively heavy load, wait approx. 30 min.
Converter help-voltage	Frequency converter auxiliary voltage	ON, when the frequency converter is switched on.	Check the frequency converter.

Status message	Explanation	Status	Hints for troubleshooting
Halt circuit	Table stop circuit output.	Open, if table stop is activated at the operating panel or intercom.	Check the cable connection to the operating panels or dummy plug (X28, on the backplane) must be present if the rear operating panel is not connected.
Temperature vertical drive	Temperature of the vertical drive A531 is too high.	ok	Reload/All. Overload for table due to non-stop movement or an excessively heavy load. Wait appr. 30 min., then Reload/All.
Voltage of intermediate circuit	Voltage is present in the intermediate circuit (A512)	Ok, if the patient table can be moved horizontally.	
Output halt flip flop	Table stop monitor	Low, no table stop. High, table stop.	
Collision detection	Collision vertical	Low	Move the table up, if it is resting on an obstacle. Activate table stop and reset. Check switches SY6, SY15 a/b, Refer also to the (→ Status messages / Page 21)
Internal error of hor. ampl.	Error in horizontal end stage A512 (over-/undervoltage, overtemperature).	ok	Reset table stop, if the error occurred in conjunction with other errors. Check the patient table supply voltage (195.5V to 253V) on A5131 pin X100.3 and 6. Replace A512.

Status message	Explanation	Status	Hints for troubleshooting
High current error of hor. ampl.	Shorted output or internal error in horizontal end stage (A512).	ok	Switch off F15 (CCA) and disconnect cable W5421 at the backplane. Switch on F15. If the LED H2 on A512 is glowing, replace A512 and reconnect cable W5421. Cable W5421 or motor A532 are shorted out. Check the cable and motor and replace, if necessary.
Antivalence error	Missing antivalence of the signals from the incremental generator of the horizontal drive (A532)	ok	Reload/All and press Home button on the operating panel. Check signal path to the incremental generator (A532). Check voltage supply (+5V; LED H1) at power supply (A5132) and at cable W5422-X2 pin 1 and 6. Replace horizontal drive A532. Replace controller A5140/A514S.
No null impulse plausibility	No zero pulse plausibility of the incremental generator of the horizontal drive (A532)	ok	Reload/All and press Home button on the operating panel. Check signal path to the incremental generator (A532). Check voltage supply (+5V; LED H1) at power supply (A5132) and at cable W5422-X2 pin 1 and 6. Replace horizontal drive A532. Replace controller A5140/A514S.
Communication to motion controller	Internal or communication error in/to motion controller A5141.	ok	Power-on-Reset via F15 (PCA). Replace the controller A5140/ A514S.

Status message	Explanation	Status	Hints for troubleshooting
Reference status	Reference pulse is out of range No reference pulse within the expected range no reference switch found	ok	Check SZ10 and the signal path to SZ10. Refer also to the (→ Status messages / Page 21). Check the belt tension.
Switch OFF state (Hor. Driver)	Switch-off error at the horizontal driver (A512).	ok	Reset the table stop. Replace horizontal driver A512. Replace controller (A5140/A514S).
Switch ON state (Hor. Driver)	Switch-on error at the horizontal driver (A512).	ok	Check the fuses /LED’s on the horizontal driver (A512). LED H5 has to glow continuously otherwise: check fuse F31 on board A512 check/replace power supply A5131, transformer T1, cable W5452. LED H4 should glow only when horiz. axis is active (goes off slowly after switching to inactive), if not check/replace F31 and F32 on horizontal driver(A512), controller (A5140/A514S).
Maximum current error	Motor current for horizontal axis has been exceeded.	ok	Check for any obstacles and remove them. Check that the table moves smoothly. Reset the table stop. Check the fuses/LED’s on the horizontal driver (A512).
Horizontal movement error	The controller detected that table movement is erratic and stopped table movement.	ok	Check for any obstacles and remove them. Check that the table moves smoothly. Reset the table stop.

Status message	Explanation	Status	Hints for troubleshooting
Switch on error of converter	Vertical end stage (frequency converter A511) could not be switched on	ok	Check fuse F7 on A5131. Check the cables to the frequency converter (W5451, W5453). Replace frequency converter A511. Replace A5131.
Switch off error of converter	Vertical end stage (frequency converter A511) could not be switched off	ok	Reload/All and press Home button on the operating panel. If the error is still present, replace: power supply A5131 controller A5140/A514S.
Converter state	Parameter change or communication error from the frequency converter	ok	Patient table in Home position, Reload/All. Reinitialize frequency converter using the (→ Mode M7, parameterizing the converter / Page 30) Replace frequency converter A511.
	Incorrect version of frequency converter firmware		Frequency converter installed is the wrong revision level; should be at least "G". Replace with a new frequency converter.
EEPROM state	Cecksum error or incorrect loadware version	ok	Reload/All (forced). Error appears together with I2C bus. Replace power supply A5132. Replace operating panel A516 or display A517.

1.3.4 Status of switches

Tab. 1 with no trolley connected

Patient table position in Z-direction		magnet center	outside (HOME) [1]	outside
Patient table position in Y-direction		**highest position**		**lowest position**
Toggle switch between vertical mode and horizontal mode	SY 0_2	on	on	off
Lowers table position (without trolley)	SY 1	on	on	off
Switches to slow speed during table upward movement	SY 2	off	off	on
Table speed switches to slow speed if the table moves down and a trolley is connected; table movement stops if the table moves up and a trolley is connected	SY 3	on	on	on
Table movement stops if the switch is activated when the table is moving down and a trolley is connected	SY 4	on	on	on
Lowers table position (with trolley)	SY 5	on	on	off
Supervision for spindle nut (SY6) and for carrier plate (SY15a/b)	SY 6_15	ok	ok	ok
Toggle switch between horizontal mode and vertical mode	SY 0_1	off	off	on
Locks the vertical mode if the table is in the horizontal mode and acts as a reference switch for the horizontal mode	SZ10	off	on	on
Senses any obstacles beneath the table	SZ 11	off	off	on
Sensor; verifies the trolley connection to the patient table	SZ 12	off	off	off
Verifies the connection between the carrier panel and the removable tabletop	SZ 13	on	on	on
Two switches in parallel; verifies the trolley connection to the carrier panel	SZ 14	on	on	on

1. the frequency converter is not activated (the converter display is off)

Tab. 2 with trolley connected

Patient table position in Z-direction		magnet center	outside (HOME) [1]	outside	
Patient table position in Y-direction		**highest position**		**trolley height [2]**	**lowest position**
Toggle switch between vertical mode and horizontal mode	SY 0_2	on	on	off	off
Lowers table position (without trolley)	SY 1	on	on	on	on
Switches to slow speed during table upward movement	SY 2	off	off	on	on
Table speed switches to slow speed if the table moves down and a trolley is connected; table movement stops if the table moves up and a trolley is connected	SY 3	on	on	off	on
Table movement stops if the switch is activated when the table is moving down and a trolley is connected	SY 4	on	on	off	on
Lowers table position (with trolley)	SY 5	on	on	on	off
Supervision for spindle nut (SY6) and for carrier plate (SY15a/b)	SY 6_15	ok	ok	ok	ok
Toggle switch between horizontal mode and vertical mode	SY 0_1	off	off	on	on
Locks the vertical mode if the table is in the horizontal mode and acts as a reference switch for the horizontal mode	SZ10	off	on	on	on
Senses any obstacles beneath the table	SZ 11	off	off	off	off
Sensor; verifies the trolley connection to the patient ta-ble	SZ 12	off	off	on	on
Verifies the connection between the carrier panel and the removable tabletop	SZ 13	on	on	on	off
Two switches in parallel; verifies the trolley connection to the carrier panel	SZ 14	on	on	off	off

1. the frequency converter is not activated (the converter display is off)
2. the trolley is connected and locked

1.4 PMU

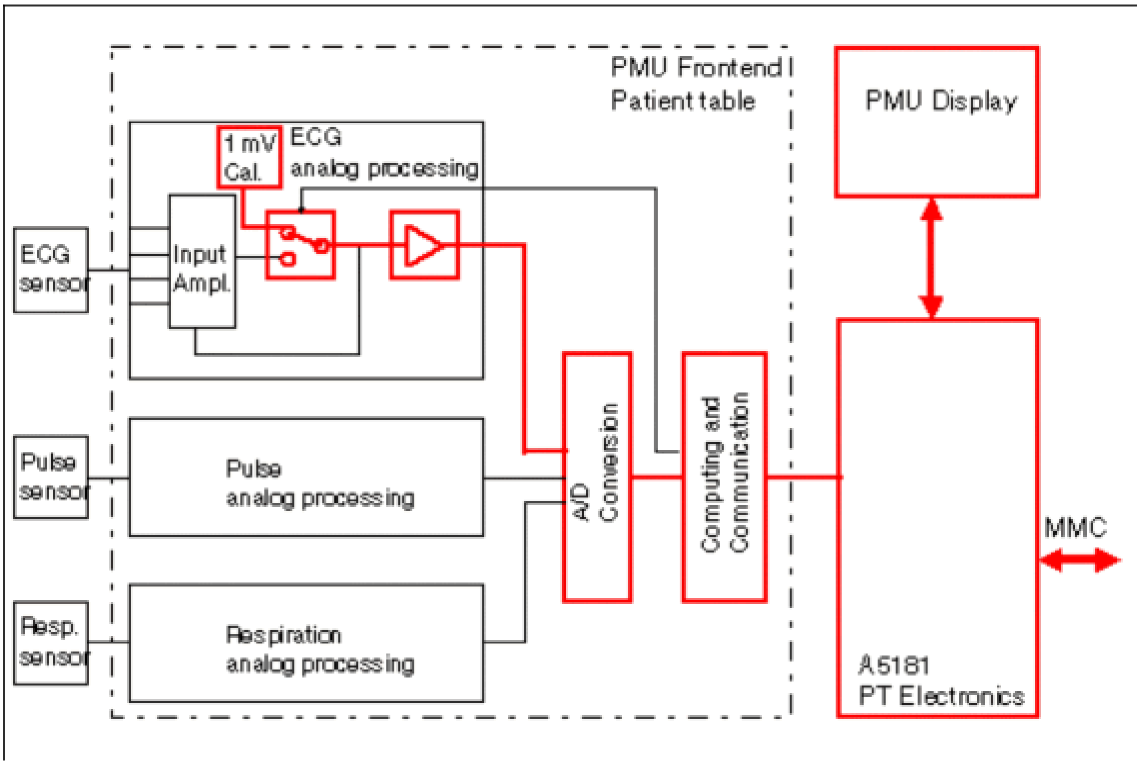
1.4.1 Status messages

Status message	Explanation	Hints for trouble shooting
Supply voltages	Monitors the supply voltages at the PMU Frontend	Check the fuse F15 on board A5132 and the fuse F16/F17 on board A5132. Check the cabling between the PMU Converter and the PMU Frontend.
DSP memory test	Tests the DSP memory in the PMU Frontend.	The PMU Frontend should be replaced.
data receiving	Verifies the communication lines of PMU Frontend.	Check the cabling between the PMU Frontend and the PMU Converter A5181 and MPCU.
Following status messages have no direct impact for service. Those are more likely application hints ECG lead(s) connected Pulse sensor connected active ECG electrodes battery voltage (of the active ECG electrodes) sensor has proper contact (of the active ECG electrodes) **Remark**: If no ECG leads or active ECG electrodes are connected, the corresponding status bits will be not o.k.		

1.4.2 PMU Transmit (interactive)

A physiological sub-menu is displayed on the PMU display at the magnet and at the physiological window. After a learn phase the calibration pulse will show up and on the PMU display at the magnet front side, in additional a beep is audible from the intercom.

Fig. 3: PMU Transmit (interactive)



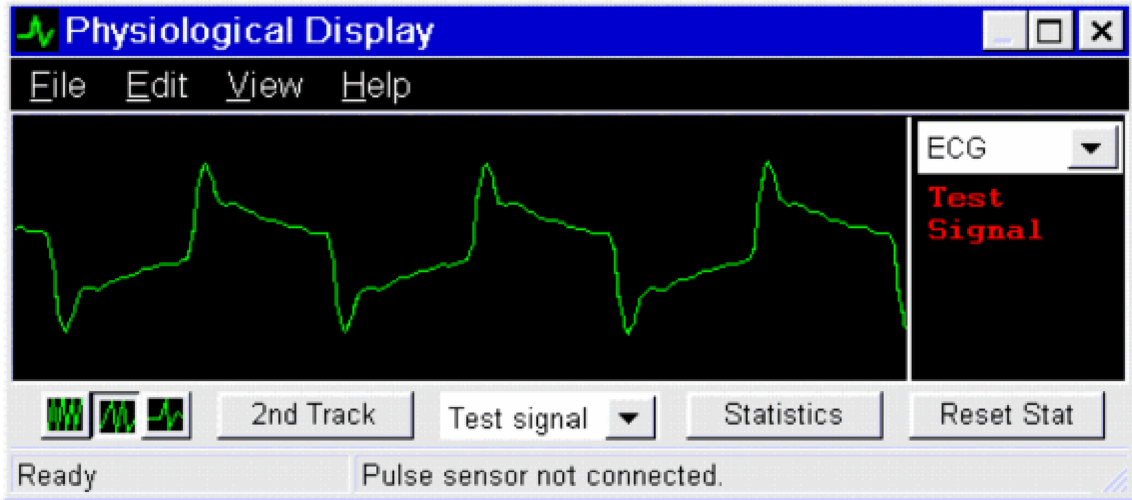
1.4.3 Signal Path

At the PMU Frontend a 1mV test signal will be generated and sent via the electr.-/optical converter to the MMC.

1.4.4 Evaluation

If both pulse and beep sound appear, the popup window should be acknowledged with OK. An example of the test signal is shown in (→ Fig. 4 Page 23).

Fig. 4: PMU signal at the physiological window



1.4.5 Conclusions

This test checks the signal path from the PMU Frontend to the MMC including the signal path to the PMU display and the intercom at the console.

1.4.6 PMU - Trouble shooting Chart

Symptom	Possible cause	Service action	Reference
No test signal is displayed at the physiological display	Loose cable connection	Check the cables between the PMU converter A5181 and the PMU Frontend. Check the fibre optical cable between the PMU Converter A5181 and the MPCU. Check the power supply +10.5/+18V/-18V for the PMU	Use the PT communication & status test
Weak ECG pulse or noise is displayed at the physiological display but a satisfactory test signal is displayed	ECG lead is defective Patient contact with the ECG electrodes is not sufficient.	Check the resistance of the ECG leads Check the proper application of the ECG electrodes (e.g. together with the application specialist, ..)	between 15k and 30kOhm
No pulse from the pulse sensor or respiratory pulse is displayed at the physiological display but a satisfactory calibration pulse is displayed	Pulse sensor or respiration belt is defective PMU Frontend defective	Check the pulse sensor on a volunteer Check the respiration belt by expanding the belt Replacement of the PMU Frontend	
“Battery low” LED on active leads is blinking	Battery in active leads is low	Replace active leads	System manual
No beep sound (test signal or ECG signal is visible) at the intercom but the intercom itself is working	The trigger switch is positioned at "0" Loose cable connection	Check the trigger switch on the left side of the intercom Check the cable between MPCU box and the Intercom	

Symptom	Possible cause	Service action	Reference
PMU Display on the magnet is off	Power supply not on	Check the F15/RF Room in the CCA Check the fuse F4 on the board A5131 Check the cable between the PMU Converter A5181 and the PMU Display	Supply voltage: appr.13VAC
No test signal is displayed on the PMU Display on the magnet, but on the physiological display on the console.	Loose cable connection	Check the fibre optic cable between the PMU Converter A5181 and PMU Display	

2.1 Patient Table

2.1.1 Service-Software for the patient table

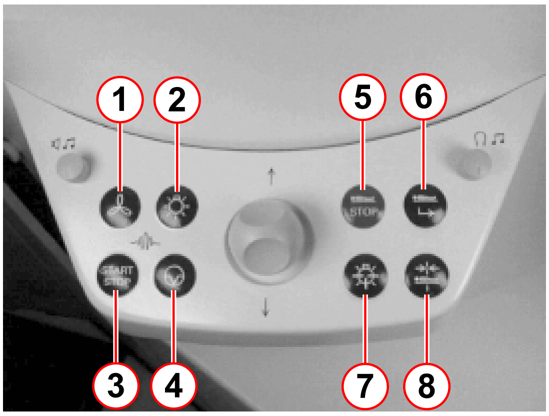
2.1.1.1 General

The service software for the patient table allows you to test and change the parameters for the patient table. If the patient table electronics is in the service mode, the table functions are limited (no table movement, very limited CAN functionality, no ventilation or lighting control, no table stop reset).

You evoke the service software by simultaneously pressing the UP and DOWN buttons on backplane A5150 (patient table electronics).

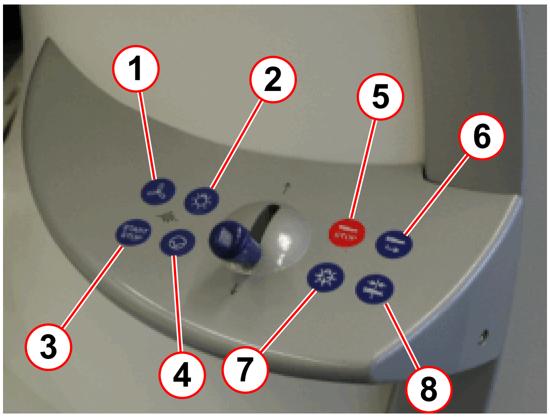
The service software for the patient table includes eight different modes, from M1 to M8. Each mode is indicated accordingly on the display together with other mode-specific information.

Fig. 5: Control panel (HaSy)



(1) patient air
(2) tunnel light
(3) start/stop measurement
(4) interrupt measurement
(5) table STOP
(6) PT out (Home)
(7) PT to isocenter
(8) laser loght localizer

Fig. 6: Control panel right (Maestro Class)



(1) patient air
(2) tunnel light
(3) start/stop measurement
(4) interrupt measurement
(5) table STOP
(6) PT out (Home)
(7) PT to isocenter
(8) laser loght localizer

2.1.1.2 How to operate the service software for the patient table

To evoke the service software:

- Press the "UP" and "DOWN" buttons on Backplane A5150
- "Start for Servicemode" will appear on the display.
- Press the “measurement start/stop” key at the control panel.
- "M1: . . ." appears on the display.

To switch modes

You can switch between modes M1 through M8 by pressing the “PT out (Home)” or “PT to isocenter” keys through the numbering sequence. The current mode (except for some display text in mode 2) is shown in the upper left of the display.

To exit from the service software

- Press the “PT to isocenter” key at the control panel until the “M8: Leave with START” appears on the display.
- Press the "start/stop measurement" key on the control panel.
- The display switches to standard mode.

2.1.1.3 Mode M1, switches and sensors

This mode shows the status of the switches and sensors, and provides an easy way to check the condition of the wiring as well as the functions of the switches and sensors. All switches and sensors are supplied via +18V_DC_SENS (board A5132). The HIGH signal means that the respective switch or sensor has been activated, that is +18V_DC_SENS is present at the controller input (board A5140).

Each switch/sensor has been assigned a single digit in the text line of the display (except for SY0 which has two digits). The reference number of the respective switch or sensor is displayed when the controller input has been set to HIGH (switch/sensor is activated). When the controller input is set to LOW, the display shows "-".

For example, the display may show: M1: SY - 0 1 - 3 4 5 6 SZ 0 1 - 3 4

Details of the display:

Switch	Ref.-No.	Signal	Description of the switch
SY 0_1	0	LOW	Toggle switch, if the table is in the vertical mode, the horizontal mode is locked
SY 0_2	0	HIGH	
SY 1	1	HIGH	Lower table position (without trolley)
SY 2	2	LOW	Switches table speed to slow during the table upward movement
SY 3	3	HIGH	If the trolley is connected and the table moves down the table speed will be switched to slow speed. If the trolley is connected and the table moves up, the table will slowly stop.
SY 4	4	HIGH	If the trolley is connected and the table moves down, the table will slowly stop.
SY 5	5	HIGH	Lower table position (with trolley).
SY 6	6	HIGH	Supervision for the spindle nut.
SZ 10	0	HIGH	Locks the vertical mode if the table is in the horizontal mode, and acts as a reference switch for the horizontal mode.
SZ 11	1	HIGH	Sensor; stops table downward movement if the trolley is not connected or if any obstacle is close to the sensor.

Switch	Ref.-No.	Signal	Description of the switch
SZ 12	2	LOW	Sensor; verifies the trolley connection to the patient table.
SZ 13	3	HIGH	Verifies the connection between the carrier panel and the removable tabletop.
SZ 14	4	HIGH	Two switches in parallel; verifies the trolley connection to the carrier panel.

- The switches for collision detection SY6, SZ15a, and SZ15b are connected in series and data is acquired in common under SY6.
- SY0 gives you the make or break contacts.

"M1: SY - 0 . . . " means that the input SY0_1 (make contact) is LOW and SY0_2 (break contact) is set to HIGH. This means, therefore, that the table is at the correct height to move into the magnet and SY0 is active.

"M1: SY 0 - . . " means that the input SY0_1 (make contact) is HIGH and SY0_2 (break contact) is set to LOW. The table is in the vertical mode.

"M1: SY - - . . " or "M1: SY 0 0 . . " are not acceptable, because they indicate errors.

2.1.1.4 Mode M2, display

In this mode, the display is provided with four different test images. Use the "measurement start/stop" key and "measurement pause" key to toggle between the images.

The contrast of the display can be changed with the keys for "funnel air" or "funnel light".

Two of the test images will show the currently selected contrast setting.

> Use the front control panel to manipulate the display in front. The same allocation applies to the control panel and display located in the back.

For example, the display may show M2: Display Test (31,31)

Explanation for the example "M2: Display Test (xx,yy)":

Contrast	for Display
xx	Magnet, in front
yy	Magnet, in back

When you exit the service software via mode M8, the contrast setting selected will be saved.

> The display should be in operation for at least 20 minutes before you correct the contrast setting.

2.1.1.5 Mode M3 through M5, control panels

These modes provide you with the status of the respective control panel, if it was recognized at the start of the patient table loadware.

- M3 operating panel PAN_0 (main operating panel, front side, magnet)
- M4 operating panel PAN_1 (auxiliary operating panel, in back of magnet)
- M5 operating panel PAN_2 (auxiliary operating panel, front side, magnet)

For example, the display may show:

M3: PAN_0 KEY: 7F JOY: 126

Key code

The information right after KEY represents the function key pressed on the control panel

Display	Button
7D	patient air
7E	tunnel light
7B	start/stop measurement
77	interrupt measurement
3F	laser light localizer

- “PT out” and “PT to isocenter” keys are tested during the mode changes.
- It is not possible to test the “table STOP” key with the service software. However, you can test its function during normal table operating mode.

Joystick code

The decimal value following JOY represents the joystick position.

Display	Joystick position
127 to 128 (ideal)	Center (not deflected)
117 to 138 (acceptable)	Center (not deflected)
000	Joystick deflected all the way down
255	Joystick deflected all the way up

The table will begin to move at values below 96 or above 160.

If the control panel is not recognized, the display shows, e.g.: "M4: PAN_1 not found"

2.1.1.6 Mode M6, position acquisition

This mode measures the horizontal table position in increments (inc) and converts them into mm’s.

The display may show, for example:

(reference position is known)

M6: Zpos: SZ10+009344inc

0024mm

The display may show, for example:

(reference position is not known)

M6: Zpos: ????+001234inc

0003mm

The position changes immediately in response to tabletop movement.

The entire motorized tabletop travel is 2160 mm or 847,830 inc, that is, 392,514 inc/mm. You can override the table travel limit manually (e.g. after pressing the “table STOP” key).

2.1.1.7 Mode M7, parameterizing the converter

In this mode, the currently valid set of parameters can be transferred to the converter.

> i Parameters can be changed for the converter only if, prior to starting the service software, the table is in a position where the frequency converter is connected (the converter display shows 000.0).

> i The converter should not have been disconnected in the meantime (e.g. by pressing the “start/stop measurement” key).

The display may show

(converter is enabled)

M7: INV-Init with START

The display may show

(converter is not enabled)

M7: INV not available

- Press the “start/stop measurement” key to start the parameterization of the converter.
- The written and stored parameter is shown on the display during automatic transfer of parameters.
- Automatic transfer of parameters is completed in approximately 35 seconds and cannot be interrupted.
- Parameterization completed successfully if the display shows “M7: Success -> CENTER”
- Exit the mode with the “PT to isocenter” key.

2.1.1.8 Mode M8, Exiting from the Service mode

- Exit the mode with the “start/stop measurement” key.

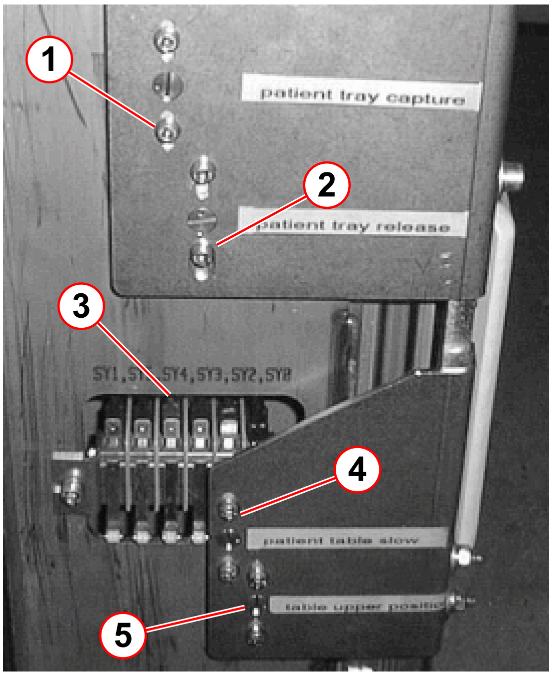
2.1.2 Switch adjustment

Switches SY0, SY... are located on the small cam plate under the left lifting column cover.

If the adjustment range is not sufficient, the small cam plate must be shifted.

The side cover will have to be removed prior to adjusting this switch. Refer to (→ Removing the column cover / MR-000.841.15).

Fig. 7: SY switch



(1) cam SY4
(2) cam SY3
(3) cam SY1, 5, 4, 3, 2, 0
(4) cam SY2
(5) cam SY0

2.1.2.1 Switch SY0/SY2

- Place a weight of approximately 80 kg on the table (use all the phantoms or a volunteer).
- Move the patient table upward and into the magnet.
- The carrier plate rollers should sit exactly on the slideway and should be stable.

Carrier plate position:

carrier plate too high	Release the cam for SY0 and **move it down** by adjusting the eccentric screw. The cam for SY2 should be moved the same distance as for SY0.
carrier plate too low	Release the cam for SY0 and **move it up** by adjusting the eccentric screw. The cam for SY2 should be moved the same distance as for SY0.

- Check the adjustment and repeat, if necessary.

2.1.2.2 Switch SY3/SY4

The docking heights for upward and downward travel are set separately.

> Detailed adjustment instructions are located in (→ Adjusting the docking height / M1-010.815.01)

- Position the trolley under the table. Activate the foot brake.
- Place a weight of approximately 80 kg on the table (use all the phantoms or a volunteer).
- Move the table down until it stops above the trolley.

Ideally, there should not be any gap between the removable table and the trolley. You must be able to lock in the tabletop using the foot lever from this position (both latches snap in without rubbing).

Carrier plate position:	
The carrier plate stops before it reaches the trolley.	Release the cam for SY4 and **move it up** by adjusting the eccentric screw. The cam for SY3 should be moved the same distance as for SY4.
The carrier plate continues to move, even though it is already on the trolley.	Release the cam for SY4 and m**ove it down** by adjusting the eccentric screw. The cam for SY3 should be moved the same distance as for SY4.

- Check the adjustment and repeat it, if necessary.

You have now finished adjusting the docking height from above. Now you can check or adjust the docking height from below.

- Lock in the carrier plate using the foot lever on the trolley.
- Move the carrier plate approximately 40mm downward.
- Move the carrier plate up as far as possible (it should stop automatically at docking height).

The carrier plate must now lie flush on the trolley, and the trolley locking mechanism must be functioning correctly.

The carrier plate should move further upward	Release the cam for SY3 and **move it down** by adjusting the eccentric screw.

- Check the adjustment and repeat it, if necessary.
- Move the carrier plate further up and remove the trolley.

Overlap of switches:

The overlap of both switches should be between 4 mm and 8 mm. Check this using the service software for the patient table (→ Mode M1, switches and sensors / Page 27).

Checking the entire docking process

As long as the transfer (with and without patient load) of the carrier plate to the trolley and back functions correctly, the adjustment is complete.

2.1.2.3 Switch SZ10

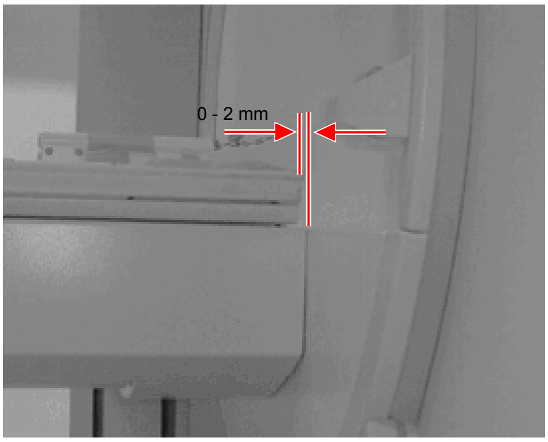
After each adjustment of SZ10 or the cam, the table must be switched briefly off, power on Reset via F15 (PCA).

Setting / checking the reference position for the basic table

- Remove the end cover and the base cover of the support arm. Switch SZ10 is located in the support frame on the magnet side.
- Move the tabletop by approximately 30 cm in the direction of the magnet and then back until it automatically stops.
- Move the table into the “Home-Position”. This initial movement of the table after switching on the voltage is the reference movement.

The head-end of the tabletop must be flush with the table carrier; there must not be any overhang.

Fig. 8: SZ10



The **space** between the tabletop and table carrier should be between 0 and 2mm.

To make the distance smaller	Release switch the mounting for switch SZ10 and move it up.

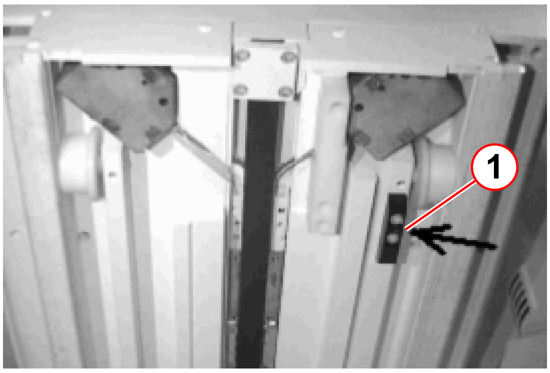
- Switch the system off again briefly and check the adjustment. Repeat the adjustment, if necessary.
- Install the end cover and the base cover of the support arm.

Adjustment/inspection of reference position for trolley operation

- Move the tabletop by approximately 30 cm in the direction of the magnet and then back until it automatically stops.

The head-end of the tabletop must be flush with the table carrier; there must not be any overhang.The space between the tabletop and table carrier should be between 0 and 2 mm.

Fig. 9: cam for SZ10



The distance must be changed.	Move the tabletop far enough into the magnet so that it reaches the switch cam underneath the tabletop carrier. Adjust the cam by the same distance and in the same direction as the deviation.

- Switch the system off again briefly and check the adjustment.

2.1.2.4 Switch SZ11

Sensor SZ11 is an optical reflex sensor. It prevents damage to the table and trolley in connection with the mirror, in case the trolley does not dock correctly and is still located beneath the table. It also senses other obstacles as well (but about 10 times worse than mirrors) and switches off downward movement. Since the sensor could interfere with the measurement, voltage to the sensor is switched off at times. In service mode, it is always switched on and can be checked.

Setting the sensitivity

- Remove the covers so that sensor SZ11 is accessible at the foot-end of the carriage frame.
- Rotate the adjustment screw (blue plastic) on the sensor to the right stop (corresponds to maximum sensitivity.
- Using the service software for the patient table (→ Mode M1, switches and sensors / Page 27) check whether sensor SZ11 switches on when an obstacle (e.g. hand) is placed in front of it.
- Perform the same check with the trolley.
- Reinstall all covers.

> **i** Do not turn the adjustment screw too far.

2.1.2.5 Switch SZ12

Sensor SZ12 comprises a reflex barrier in conjunction with a small mirror in the docking terminal. It detects whether a trolley is correctly docked. Since the sensor could interfere with the measurement, the supply voltage is switched off for part of the time. In service mode, the sensor is always switched on and can be checked.

Adjusting the sensitivity

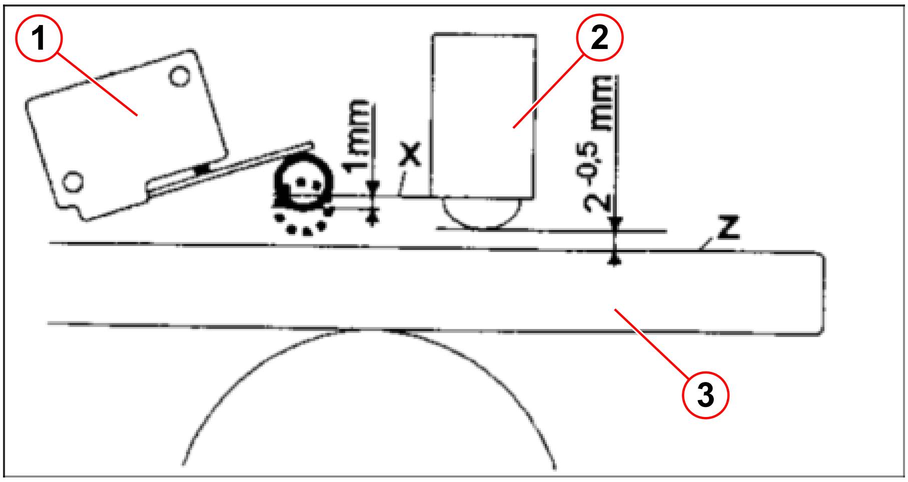
- Remove the covers so that switch SZ12 is accessible.
- Turn the adjustment screw (blue plastic) on the sensor to the right until it reaches the end stop (corresponds to maximum sensitivity).
- Position the trolley so that the vertical guide rod is 4 mm before the end stop.
- Shift the sensor laterally so that it switches on 4 mm before the guide rod reaches the end of the docking terminal. Check the adjustment using the service software for the patient table (→ Mode M1, switches and sensors / Page 27).
- Reinstall the covers.

2.1.2.6 Switch SZ15a/b

The spring-loaded cylinder prevents switch SZ15a/b from actuating at too low a pressure. The switch is only actuated when the ball actuator is pressed down into the cylinder.

- Remove the covers so that switch SZ15 a/b is accessible.

Fig. 10: SZ15



(1) SZ15
(2) Spring-loaded cylinder
(3) carrier panel

- Adjust the switch according to the instructions above.
- Pull the carrier panel slowly upward at the foot-end, first on one side (then on the other). Check the switching action of switch SZ15 a/b using the service software for the patient table (→ Mode M1, switches and sensors / Page 27).

> Check the display for SY6, since switches SZ15 and SY6 are connected in series.

- Reinstall the covers.

2.1.3 Description of LED's

Reference	Name	Function
Power supply		
A5132-H1	**+5V_DC_ENC**	power on
A5132-H2	**+7.5V_DC_CON**	power on
A5132-H3	**+18V_DC_SENS**	power on
A5132-H4	**+10.5V_DC_PAN**	power on
A5132-H5	**+10.5V_DC_PMU**	power on
A5132-H6	**+18V_DC_PMU**	power on
A5132-H7	**-18V_DC_PMU**	power on
A5132-H8	**+13.5V_DC_VID**	power on
A5132-H9	**+18V_DC_TMP**	power on
A5132-H10	**-18V_DC_TMP**	power on
Lighting		
A5182-H1	**POWER** or U=	Power supply (intermediate circuit)
A5182-H2	**LAMP_1** or UL1	Voltage drop in lamp 1 (on the right when looking at the back of the magnet) ▪ brightness according to light level ▪ off if the lamp shorts out (or 0 level is selected)
A5182-H3	**LAMP_2** or UL2	Voltage drop in lamp 2 (on the left when looking at the back of the magnet) ▪ brightness according to light level ▪ off if the lamp shorts out (or 0 level is selected)
A5182-H4	**LSB_LEVEL** or 2^0	Nominal value for light level (LSB, 2^0)
A5182-H5	**MSB-LEVEL** or 2^1	Nominal value for light level (MSB, 2^1)
A5182-H6	**TEMP_FLT**	Overtemperature of heatsink
End stage A512		
A512-H5	**POWER**	38 V_AC - Power supply for end stage
A512-H4	**SUPPLY**	Power supply (Intermediate circuit) ▪ off during MR measurement (glows for approx. 30 seconds afterward

Reference	Name	Function
A512-H3	**RUN**	End stage operating ▪ switched on and enabled and no end stage error
A512-H2	**ERROR_C**	Short on output circuit (Motor A532, cable W5421, end stage A512)
A512-H1	**ERROR_S**	Summary error message ▪ on for over or under voltage or overtemperature of the end stage
Controller		
A514-H1	**FW**	FW LED of the piggy board ▪ flashes, when FW is active ▪ glows (briefly) when parameters are being saved in the EEPROM
A514-H2	**LW**	LW LED of the piggy board ▪ flashes when the LW is active ▪ on during sleep mode
A514-H3	**HORIZ FAULT**	Horizontal axis error ▪ on in (Re-) start phase or for Power-On (for approx. 1s) ▪ on for horizontal axis error If the loadware (e.g. invalid) could not initialize the position controller A5141, the controller does not report a horizontal error. ▪ on during sleep mode
A514-H4	**HORIZ OK**	Horizontal axis ready ▪ on, if there is no horizontal axis error (refer to HORIZ FAULT) ▪ on during sleep mode

2.1.4 Description of fuses

Assembly Fuse	Value	Name of line	Assembly Fuse	Test points	Units protected
A5131-**F1**	3,2AT	16.5V_AC_1			Switches, sensors, brake, PMU Front end, Gradient coil monitor (+18V_DC respectively)
A5131-**F2**	1,6 AT	16.5V_AC_2			PMU Front end, Gradient coil monitor (-18V_DC respectively)
A5131-**F3**	6,25 AT	10V_AC			Controller, incremental encoder, operating panel, displays, light localizer, PMU Front end and PMU converter
A5131-**F4**	3,2 AT	13.5V_AC			PMU display
A5131-**F5**	3,2 AT	-		A5131-X4.1 A5131-X4.2 ! 230 VAC !	Power line transformer A5131-T1, primary
A5131-**F6**	1 AT	-			Patient air
A5131-**F7**	15 AM	-			Frequency converter A511
		GND		A5132-X12	
A5132-**F11**	0,8 AT	+18V_DC_SENS	A5132-H3	A5132-X4	Microswitch/sensors SYx and SZxx, SZ13 (incorporated in the ASV) Brake for drive A532
A5132-**F12**	0,8 AT	+13.5V_DC_VID	A5132-H8	A5132-X9	Video monitor camera
A5132-**F13**	3,2 AT	+10.5V_DC_PAN	A5132-H4	A5132-X5	Operating elements, displays and light localizer
A5132-**F14**	3,2 AT	+7.5V_DC_CON +5V_DC_ENC	A5132-H2 A5132-H1	A5132-X3 A5132-X2	Controller A514 Incremental encoder of drive A532
A5132-**F15**	0,8 AT	+10.5V_DC_PMU	A5132-H5	A5132-X6	PMU converter A5181, PMU radio
A5132-**F16**	0,8 AT	+18V_DC_PMU	A5132-H6	A5132-X7	PMU Front end
A5132-**F17**	0,8 AT	-18V_DC_PMU	A5132-H7	A5132-X8	PMU Front end
A5132-**F18**	0,2 AT	+18V_DC_TMP	A5132-H9	A5132-X10	Gradient coil monitor D9

Assembly Fuse	Value	Name of line	Assembly Fuse	Test points	Units protected
A5132-**F19**	0,2 AT	-18V_DC_TMP	A5132-H10	A5132-X11	Gradient coil monitor D9
A5182-**F21**	4 AT	18V_AC	A5182-H1	-	Tunnel light
A512-**F31**	4 AT	38V_AC	A512-H5	-	Horizontal end stage A512
A512-**F32**	4 AT	-	A5132-H4	-	DC intermediate circuit for the horizontal end stage A512

2.2 Video Monitoring

2.2.1 General

This section of the TSG deals with problems regarding the video monitoring. The strategies and procedures provide the necessary information for servicing the video monitoring.

The primary components of the patient video monitoring are:

Patient CCD camera

Patient Video Display

CAUTION

Strong magnetic field.

Magnetizable tools or objects introduced into the magnetic field become dangerous projectiles.

» **Use only non-magnetizable tools and do not carry magnetizable objects when working within the magnetic field. Use utmost caution to avoid hazardous conditions with regard to equipment and personnel.**

2.2.2 Additionally required documents

Function Description "Patient Handling" chapter

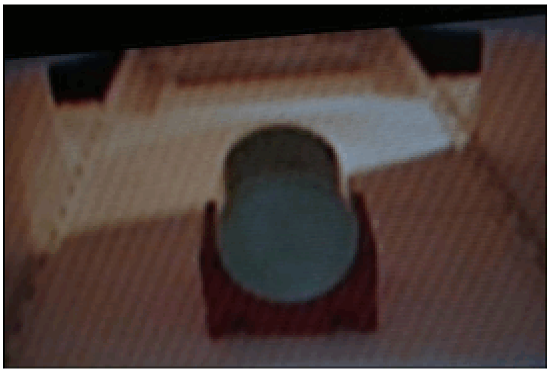
Replacement of Parts "Patient Handling" chapter

2.2.3 Common problems

Moire structure or lines with different intensity, frequency and orientation on the patient video display:

Due to the magnetic field, the CCD camera may generate a moire structure or lines on the patient video display (→ Fig. 11 Page 41). To check that the CCD camera is causing the moire structure or lines on the patient video display, remove at the rear of the magnet the upper center camera cover and dismount the camera. Move the CCD camera approx. 30cm away from the magnet and check if the moire structure on the patient video display is gone (a second person may be needed). In case the moire structure is gone after positioning the camera 30cm away from the high field area, only a replacement of the CCD camera will help. A very low intensity moire structure, barely visible, does not indicate a problem, if this is the case the CCD camera should not be replaced.

Fig. 11: Strong moire structure on patient video display



» The common print number (MR-000.840.01) from the Troubleshooting Guide is divided into separate Component print numbers.

- Chapter Video Monitoring has been added.

Version 02

- Test Point A5132 - X3 to A5132 -X2 changed.

There are no Hazard IDs in this document.

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